

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTESC304 - COMPUTER ARCHITECTURE & OPERATING SYSTEMS

Time : 3-Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The fundamental unit of data in a computer is:
 - a) Byte
 - b) Bit
 - c) Word
 - d) Nibble
- GPGPU stands for:
 - a) General-Purpose Graphics Processing Unit
 - b) General-Purpose Graphical Processing Unit
 - c) Great Performance Graphics Processor Unit
 - d) General Performance Graphic Processing Unit
- 3 What is the main difference between RISC and CISC architecture?
 - a) Clock speed
 - b) Number of registers
 - c) Number of instructions
 - d) Operating voltage
- RAID stands for:
 - a) Random Access Independent Disks
 - b) Redundant Array of Inexpensive Disks
 - c) Repeated Access Integrated Devices
 - d) None of the above
- MESI protocol is used in:
 - a) Virtual memory
 - b) Cache coherence
 - c) Page replacement
 - d) Instruction scheduling
- 6 Which of the following is not a function of an operating system?
 - a) Process management
 - b) File management
 - c) Data mining
 - d) Memory management

- 7 Which component performs arithmetic and logic operations?
 a) Control Unit
 b) ALU
 c) Registers
 d) Cache
- 8 Which format is used to represent floating point numbers in computers?
 a) IEEE 754
 b) ASCII
 c) BCD
 d) UTF-8
- 9 Which memory type allows simultaneous read and write operations?
 a) RAM
 b) ROM
 c) PROM
 d) EPROM
- 10 Hardwired control unit is typically:
 a) Faster but less flexible
 b) Slower and flexible
 c) Always microprogrammed
 d) Only used in GPUs
- 11 The part of OS that loads first into memory is:
 a) Shell
 b) Kernel
 c) Scheduler
 d) File System
- 12 In paging, each process is divided into:
 a) Pages
 b) Blocks
 c) Sectors
 d) Nodes

Q2. Solve the following.

- A) Describe the structure and function of a CPU with the help of a labeled diagram. 6
 B) Discuss the types of operations and operands supported by instruction sets. 6

Q3. Solve the following.

1. Describe the organization and working of semiconductor main memory with a diagram. 6
 2. Describe the process of memory allocation using:
 1. Continuous memory allocation
 2. Fixed and variable partitions
 3. Fragmentation and compaction 6

Q4. Solve Any Two of the following.

- A) Explain the steps involved in micro-instruction sequencing and execution. 6
 B) Write a short note on multiple processor organization and its types. 6

- C) Compare and contrast: 6
1. Programmed I/O
 2. Interrupt-driven I/O
 3. DMA
- Q5. Solve Any Two of the following.
- A) Explain the types of Operating Systems with examples. Also discuss the structure of an Operating System. 6
- B) Explain multiple processor scheduling and real-time scheduling with examples. 6
- Write a detailed note on System Design and Implementation of Operating Systems. 6
- Q6. Solve Any Two of the following.
- What is Process Synchronization? Explain the Critical Section Problem and outline software solutions for two or more processes. 6
- B) Write detailed notes on the following: 6
1. Semaphores
 2. Monitors
- C) Discuss in detail the methods for handling deadlocks. 6

*** End ***